

Experiment 3.2

Determination and plotting of Force experienced by a charge with varying magnitude and distance

Code File Name: Exp3_2_ChargeForce_DifferentMagnitudes.m

Code

```
clc;  
q1 = 5e-6;  
q2 = 2e-6;  
epsilon = 8.854e-12;  
k = 1/(4*pi*epsilon);  
r = linspace(0.5,5,100);  
F = k*q1*q2./(r.^2);  
plot(r,F,'LineWidth',2);  
xlabel('Distance (m)');  
ylabel('Force (N)');  
title('Force between two charges (different magnitudes)');
```

Output Graph

